

Secure QR Code Development

Problem Statement:

QR Codes that sit in the public domain, can be interfered with.

The technology around QR codes (**ISO/IEC 18004**) cannot be ignored, as it is a convenient and fast way to transfer information from a screen or surface, billboard or advertisement, to a user device for people to seek further information. The QR code is versatile, with different data storage options available, and has built-in error-correction.

But it is inherently insecure, where the physically printed QR code can be interfered with, or replaced.

Objective:

- Building an authenticated info system for QR codes.
- Building a management system for authenticated QR codes
 - Limited expiry timeout
 - Unique auth code
 - Redirect system
 - Telemetry and Metrics about a QR code usage.
- Building a mobile application to scan QR codes, and authenticate them.
- Application & secure QR code system must be backward compatible with “regular QR” code scanners.^{^1}

Expected Outcomes:

- QR code generator system & management platform
- Redirect system
- Metrics – system usage
- Alerts – from mobile application DETECTING INTERFERENCE
 - Alert should contain Location (using mobile device information)
 - Reporter name (if user wishes to share)
 - Contact information (if user wishes to share)
 - Photo of QR code and surroundings.

- Description of device or location

Optional Extras:

- Extend ISO 18004 standard to incorporate above methodology
- Time base redirect service – for time based applications or systems.

*1 – backward compatibles means: a regulation QR code scanning software system should still be able to scan and recognise regular TEXT or URL coded information. Any security or hashing added to the encoded text mustn't interfere with a "regular QR code scanner" – ie built into Apple or Google. This is to ensure wider adoption of a secure QR code standard, by assuring backwards compatibility with older/insecure QR code scanning tools.